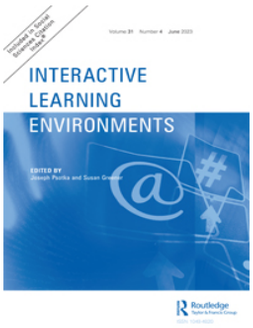

Interactive Learning Environments

ISSN: 1049-4820 (Print) 1744-5191 (Online) Journal homepage: www.tandfonline.com/journals/nile20

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To cite this article: Iolie Nicolaidou, Petros Pissas & Dimitrios Boglou (2023) Comparing immersive Virtual Reality to mobile applications in foreign language learning in higher education: a quasi-experiment, Interactive Learning Environments, 31:4, 2001-2015, DOI: 10.1080/10494820.2020.1870504

To link to this article: https://doi.org/10.1080/10494820.2020.1870504

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INTERACTIVE LEARNING ENVIRONMENTS
2023, VOL. 31, NO. 4, 2001–2015
https://doi.org/10.1080/10494820.2020.1870504

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Comparing immersive Virtual Reality to mobile applications in foreign language learning in higher education: a quasi-experiment

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ABSTRACT

Virtual Reality applications are predicted to create a paradigm shift in education, but there is little empirical evidence of their educational value. Virtual Reality affordances to support language learning have not yet been realized. This quasi-experimental study investigated the effect of a Virtual Reality application on foreign language learning and compared engagement, engrossment, and immersion between two versions of the application (Virtual Reality and mobile). Twenty experimental group undergraduate students used a head-mounted Virtual Reality display and twenty control group students used a mobile application for learning Italian as a foreign language. Data sources included: (a) a 10 open-ended questions pre-test and post-test measuring vocabulary skills, and (b) a validated questionnaire with 21 seven-point Likert scale items measuring engagement, engrossment, and immersion. Findings revealed a statistically significant difference in the experimental group students' vocabulary performance when students' pre-test and post-test scores were compared. Engagement, engrossment, and immersion received relatively high scores by Virtual Reality participants. The study did not find a statistically significant difference between the two conditions. The study provides preliminary empirical data indicating that Virtual Reality applications can be both effective and engaging for language learning, even though they were not found to be superior to mobile applications.

ARTICLE HISTORY
Received 26 August 2020
Accepted 21 December 2020

KEYWORDS
Virtual reality applications; language learning; higher education; immersion; engagement; engrossment

Introduction

Virtual Reality (VR) applications are predicted to create a paradigm shift in education and training, but there is little empirical evidence of their educational value (Makransky et al., 2019). VR is defined as a system that aims to bring simulated real-life experiences, providing topography, movement, and physics that offer the illusion of being there (Lin & Lan, 2015). VR systems can be classified into three categories: immersive, semi-immersive, and non-immersive (Ma & Zheng, 2011). Immersive VR systems are the ones in which the user wears a head display unit which includes a speaker and interacts with the system using data gloves or bodysuits, hand-held sensors, or cameras. In non-immersive VR, the user is placed in a three-dimensional environment that can be directly manipulated with a conventional graphics workstation using a monitor, a keyboard, and a

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mouse. A semi-immersive VR system comprises of a relatively high-performance graphics computing system coupled with either a large screen monitor, or a large screen projector, or multiple television projection systems (Ma & Zheng, 2011). According to Kaplan-Rakowski and Wojdynski (2018) "unlike limited Virtual Reality prototypes, the new wave of Virtual Reality headsets (especially Oculus and HTC Vive) is creating settings for immersive learning at an affordable cost" (p.125), which partly explains Virtual Reality's increasing popularity. Educational tools that have traditionally been used on a desktop computer (or mobile devices) are more recently offered in immersive VR versions involving head-mounted displays, with the expectation that a higher level of immersion will increase student motivation and learning (Makransky et al., 2019).

VR is advocated as a promising arena for language learning because it can provide an immersive and authentic environment to socially interact with native speakers (Lin & Lan, 2015). Moreover, studies have shown that VR may increase learner autonomy and self-efficacy, reduce learning anxiety, and foster creativity (Lin & Lan, 2015). According to Xie et al. (2019), since the VR tools can simulate an illusion of presence, one's mind could perceive the projected physical stimuli being close-by and therefore urge the brain to activate the schema in physical and motor channels, resulting in an enhanced learning experience.

VR is still reported to be one of the least published research topics in the technology-based learning field (Lin & Lan, 2015). Lin and Lan (2015) investigated the research trends in language learning in a VR environment by conducting a content analysis of findings published in the literature from 2004 to 2013 in four top-ranked computer-assisted language learning journals. They found that only 3.6% of the articles published in the four journals from 2004 to 2013 were related to the field of VR. They, therefore, argued that studies that analyze the appropriateness, application, and practices of VR, and its influence on language education were urgently needed (Lin & Lan, 2015). Kaplan-Rakowski and Wojdynski (2018) also agreed that investigations in virtual reality assisted language learning were still limited. More recently, Parmaxi (2020) conducted a systematic review to describe the current state of the art in VR empirical research in the field of language learning from 2015 to 2018. She included 17 high-impact journals and conferences in the fields of Computer-Assisted Language Learning and Educational Technology as sources and she retrieved 26 scholarly manuscripts focusing on VR from 2015 to 2018. Even though these numbers potentially indicate an increased interest in the field, there seems to be a lack of studies on fully immersive VR systems as opposed to non-immersive or semi-immersive VR systems, according to Parmaxi (2020). Fully immersive VR can be more efficient in terms of learning and richer in terms of users' experiences compared to non-immersive or semi-immersive VR systems. Parmaxi (2020) also identified a clear need for more quantitative studies for understanding the effect of the use of VR.

To address the need for quantitative studies on fully immersive VR systems in language learning that was identified in the literature, the present study focused on the immersive VR version of an application (Mondly.com), to examine whether there are learning benefits concerning its use when Greek native speakers attempt to learn Italian as a second language and to compare it with the mobile version of the same application. The VR version of the application (Mondly.com) requires a head-mounted VR display while the mobile application requires a mobile device such as a smartphone or a tablet. Mondly is a language learning application with speech recognition, which is currently offered in both Augmented Reality and Virtual Reality. According to Na-Young et al. (2019), who reviewed different types of intelligent chatbots in terms of language learning, including Mondly, this application offers a more in-depth language learning experience with a chatbot that lets users practice languages. A chatbot is a software system that can chat with a human user using a natural language (Na-Young et al., 2019). Chatbots, as conversational agents, provide a natural language interface to their users, which allows human-like interaction with any application that can simulate a conversation using artificial intelligence (AI). Chatbots offer a similar communication experience as one might experience communicating with a human user. Because their main aim is to mimic human conversations, chatbots imitate human speech patterns and convince users that they are talking to a real human while they are talking to machines (Na-Young et al., 2019).

Concerning challenges that students face when attempting to learn new vocabulary in a second language, such as Italian, which was chosen in the context of this study, Danesi (2016) states that one of the prevalent challenges is that students tend to "think" in their native language (L1) as they "speak" Italian (L2) as a second language and this guides their choice of words and other structures in the formation of sentences and utterances. Previous attempts in the literature to address this problem and alleviate learners' difficulties when learning vocabulary skills in Italian as a second language refer to the use of computer-assisted language learning systems, language learning environments, dialogue systems, and chatbots (Shawar, 2017), computer-assisted language learning games developed specifically for Italian (Zipf, 2019), and story-based techniques to help beginner learners of Italian to memorize words in a story (Shawar, 2017).

Literature review

With the advent of mobile technologies, new paradigms of teaching and learning that are technologically supported and enhanced had emerged, an example of which is mobile learning (Chee et al., 2017). Mobile learning has affordances such as portability, immediacy, connectivity, ubiquity, and adaptability and it has been studied extensively in the last decade. According to the meta-analysis conducted by Chee et al. (2017) who studied 144 refereed journal articles to review mobile learning trends from 2010 to 2015, most studies on mobile learning focused on higher education (36.17%), they were conducted in the subject area of Language Arts (12.93%), with the use of smartphones (14.09%) and they employed quantitative methods in their research design (47.92%). It seems that most studies on mobile learning focus on effectiveness, as shown by a metanalysis including 164 studies conducted from 2003 to 2010 (Wu et al., 2012). Specifically, a moderate mean effect size of 0.523 was found for the application of mobile devices to education, according to the meta-analysis conducted by Sung et al. (2016), which included 110 experimental and quasi-experimental journal articles published between 1993 and 2013.

The advantages of mobile learning have led to the development of increasingly high numbers of mobile applications that are easily available to smartphone users in many different domains, including learning about health (Tozzi et al., 2018) and language learning (Gangaiamaran & Pasupathi, 2017; Kallinikou & Nicolaidou, 2019). The rapid increase in the adoption of mobile technology for language teaching and learning, which provides easy access for any learner without the constraints of both place and time (Gangaiamaran & Pasupathi, 2017) has led to an increased interest in researching the effectiveness of mobile apps in language learning (Makoe & Shandu, 2018). A few examples of recent studies showed that mobile apps can be effective for learning vocabulary in a foreign language (Chen et al., 2019; Klimova, 2019). Other recent studies showed their motivational value, even though they did not demonstrate their superiority when compared with traditional instruction (Elaish et al., 2019). This study contributes to this effort as the effectiveness of a mobile app in language learning for vocabulary skills is examined when this is compared to an immersive VR application with the same learning goal. Immersive Virtual Reality applications, including Virtual Reality games, have been found to be motivating in recent studies of the past five years, in different disciplines, such as learning science (Alrehaili & Al Osman, 2019; Makransky et al., 2019; Parong & Mayer, 2018), chemistry (Winkelmann et al., 2019), math (Shi et al., 2019), history (Villena Taranilla et al., 2019), nursing (Chang et al., 2019) and intracultural awareness (Yeh et al., 2020). At the same time, immersive VR applications have been found to be distractive for the learners due to a higher cognitive load (Lin et al., 2019; Makransky et al., 2019). VR applications have not been found to be superior compared to traditional instructional media in science (Alrehaili & Al Osman, 2019). VR applications have also not been found to be superior to traditional instructional media when the goal was transferring well-structured knowledge (Zhang et al., 2020), declarative knowledge (Webster, 2016), and increasing science self-efficacy (Huang, 2019). In Makransky et al.'s (2019) study, for example, immersive VR simulations on head-mounted displays have been found to provide less opportunity for undergraduate students to build learning outcomes in science when compared with desktop simulations even though

students reported being more present in the VR condition (Makransky et al., 2019). Similarly, the findings of the study of Parong and Mayer (2018) showed that a PowerPoint slideshow may be more effective for teaching scientific information than an equivalent lesson in an immersive VR environment, but at the same time it may be less enjoyable and motivating for students. Contradictory findings were reported by other studies that documented a statistically significant difference in favor of the VR condition in both academic performance and motivation, in disciplines such as history (Villena Taranilla et al., 2019) and math (Shi et al., 2019).

The literature that specifically focused on the use of VR applications in foreign language learning was reviewed for the needs of this study. Several recent studies examined how VR can help in the improvement of participants' performance when learning a foreign language (Cheng et al., 2017) or in increasing participants' motivation when learning a foreign language (Kaplan-Rakowski & Wojdynski, 2018; Xie et al., 2019). Other studies focused on both aspects (Berns et al., 2011; Chung, 2012). It is important to note that while previous studies, conducted approximately a decade ago, implemented less advanced, non-immersive, or semi-immersive VR environments and platforms, such as Second Life, more recent studies, conducted between 2017 and 2020, focused on immersive VR systems and VR games. The most recent of these studies will be presented in chronological order in more detail, next.

Using immersive VR games, Cheng et al. (2017) adapted a 3D video game for learning Japanese, so that it could be played in VR with the Oculus Rift to teach embodied cultural interaction, such as bowing in Japanese greetings. Sixty-eight participants first completed a pre-test for Japanese vocabulary proficiency. The pre-test involved matching eight Japanese words to their English definitions. Players were randomly assigned to play either the Virtual Reality or non-Virtual Reality version. Participants were given basic instructions on how to play the game, such as how to interact with in-game characters by clicking on them. Each play session took approximately 15–20 min. Participants then completed a vocabulary post-test that was identical to the pre-test except for item order. Results showed that the VR design trained players how and when to bow, and that it increased participants' sense of involvement in Japanese culture. However, the impact of the VR game on learning vocabulary was inconclusive. The number of words learned by participants in the VR condition (M = 4.77, SD = 2.77) turned out to be lower than the number of words learned by participants in the Non-Virtual Reality condition (M = 5.39, SD = 2.56), but the difference was not statistically significant.

Xie et al. (2019) in their case-study explored undergraduate students' use of interactive VR tools (Google Cardboard and Expeditions) for learning Chinese as a foreign language. Twelve students were paired and role-played as virtual tour guides for six locations throughout a semester. One of the themes that emerged from qualitative data analysis was that contextualized presentations equipped participants with a more sophisticated vocabulary, even though technical difficulties persisted. A strong asset of this study was the fact that VR was examined for a whole semester and it was not a one-time experience for the learners. However, the study only included qualitative data, and findings were based on observations and participants' reflections. Therefore, the study "did not provide empirical evidence about whether these VR tools could improve students' oral proficiency in various aspects, such as pronunciation, fluency, grammar, and vocabulary, or reduce their anxiety, with a quantitative, experimental design" (Xie et al., 2019, p. 257).

A small number of recent studies used the Mondly VR application to investigate its effect on foreign language learning. One of them was the study of Kaplan-Rakowski and Wojdynski (2018), who conducted a pilot study examining students' attitudes towards Virtual Reality assisted language learning. Twenty-two participants interacted with a VR lesson and then completed a post-experience survey, which showed that students' experiences were highly immersive. Their experiment took place in a computer laboratory using the Mondly application with the Oculus Go headset. The results of the study provided preliminary evidence that VR technology is engaging and immersive for learners. However, this study did not measure students' learning as it only focused on attitudes and it did not use a standardized instrument for the measurement of immersion. Another study that investigated Mondly VR was the study of Tai et al. (2020), which compared two versions: one that was played on head mounted displays and another that was watched without VR on a regular computer

screen. The participants of the study were forty-nine seventh graders in Taiwan. Tai et al. (2020) showed that the VR players' vocabulary learning and retention was significantly higher than the control group.

Recent studies that implemented immersive VR settings in foreign language learning had inconclusive results concerning the effect of VR on learning. Tai et al. (2020) found VR to be superior to the non-VR condition for learning vocabulary in a foreign language, Cheng et al. (2017) did not find VR to be superior to the non-VR condition for learning vocabulary in a foreign language but the finding was not statistically significant, other studies did not provide empirical evidence about whether VR tools can improve students' performance in various aspects, including vocabulary (Xie et al., 2019), and other studies focused on attitudes and did not measure learning (Kaplan-Rakowski & Wojdynski, 2018). Moreover, according to the study of Tai et al. (2020), which showed a superiority of the immersive VR when this was compared with a desktop application, VR research has tended to pay more attention to desktop-based VR than to VR via head mounted displays, leaving the potentials of VR through head mounted displays, yet to be investigated. The present study attempted to address these gaps. The main objectives of our study were to investigate the effect of a VR application on foreign language learning focusing on vocabulary acquisition using a quasi-experimental pre-test post-test design and to compare engagement, engrossment, and total immersion between two versions of the application (VR and mobile).

The research questions of this study were the following:

(1) To what extent does an immersive VR application affect the foreign language vocabulary learning of undergraduate students?
(2) How does an immersive VR application compare with a mobile application concerning its effect on undergraduate students' vocabulary acquisition in a foreign language?
(3) How do students evaluate their immersive VR learning experience concerning their engagement, engrossment, and total immersion?
(4) How does the evaluation of an immersive VR application compare with the evaluation of a mobile application?

Methodology

Participants

The participants of the study were 40 (15 male, 25 female) Greek-speaking undergraduate students ($M = 19.6$ years-old, SD = 1.58) with little or no knowledge of the Italian language. Specifically, 50% (20/40) of participants had no prior knowledge of Italian, 22.5% (9/40) knew a few words of Italian, 5% (2/40) had taken a lesson in Italian in the past and 22.5% (9/40) were taking their first course in Italian during the semester that the study was conducted (spring semester 2020). The majority of students were first-year undergraduate students (65%, 26/40), 20% (8/40) were third-year students, 12.5% (5/40) were fourth-year students and the rest of them (2.5%, 1/40) were second-year students.

Description of study

The study implemented a quasi-experimental, pre-test, post-test, control group design. Four intact groups of students were invited to participate in the study. Two of the groups were randomly assigned to the experimental condition (VR application) and the other two groups were assigned to the control group condition (mobile application). Students were randomly selected from all groups until the study reached 20 participants for each condition. Twenty undergraduate students of the experimental group played the first three introductory scenarios of the application using a head-mounted VR display (Oculus Rift) for learning Italian as a foreign language (Figure 1). Twenty students of the control group used the mobile version of the same application (Figure 2).

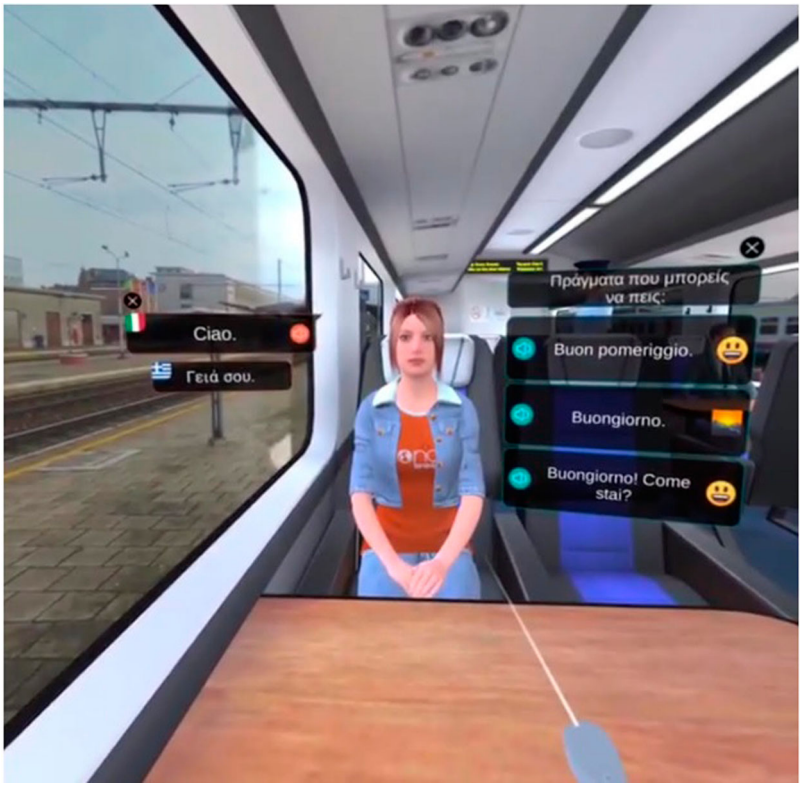

Figure 1. Screenshot of the interface of the MondlyVR application (Mondly.com) Source: https://www.mondly.com/ [Re-printed with permission by Mondly].

Mondly uses word repetition as a learning method. It allows users to memorize words and sentences in previous conversations and helps learners' pronunciation by exposing them to conversations between native speakers. With everyday scenarios such as ordering food in a restaurant, users can understand the nature of the language (Na-Young et al., 2019). The three introductory scenarios that were used in this study involve the user interacting with a taxi-driver in a taxi, with a receptionist in a hotel, and with a waiter in a restaurant. The users' native language was preselected as Greek and the users' targeted language in this study was Italian.

Data sources

Data sources included: (a) a test measuring vocabulary skills and (b) a questionnaire measuring immersion (see Appendix). The first data source was a test comprised of 10 open-ended questions measuring vocabulary skills, which was administered pre and post. An Italian native speaker confirmed that the questions were written correctly before administering the tests. The test that measured vocabulary skills was based on open-ended questions which were part of the learning material used in the first three introductory scenarios within Mondly. Participants had to answer these questions in writing, in Italian. Examples of questions are the following: "Da dove vieni?", "Che tipo di camera vuole?", "Cosa desidera?".

The second data source was a validated questionnaire with 21-items, on a seven-point Likert-type scale, measuring three factors: engagement (8 items), engrossment (6 items), and immersion (7 items) (Georgiou & Kyza, 2017). This instrument was based on a multi-leveled model of immersion with multidimensionality in each level (Georgiou & Kyza, 2017) and it can be used by researchers and designers in the field of location-based augmented reality, but it is also applicable in the field

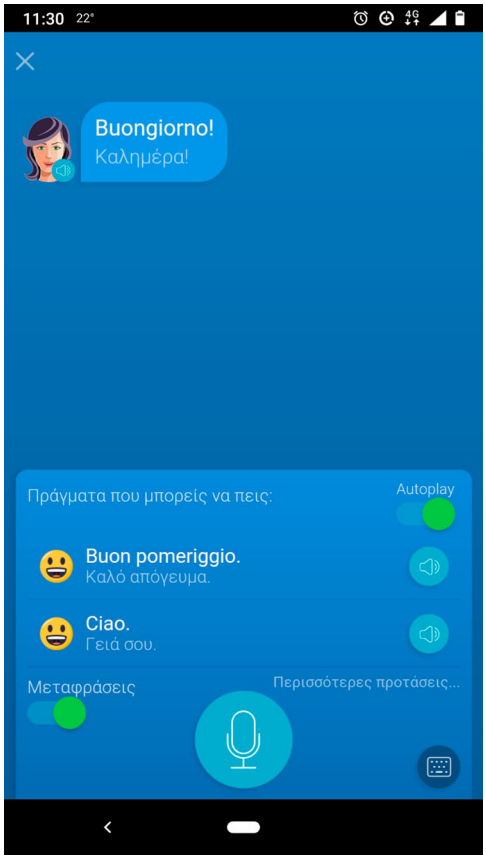

Figure 2. Screenshot of the interface of the Mondly mobile application (Mondly.com) Source: https://www.mondly.com/ [Reprinted with permission by Mondly.com].

of immersive VR learning environments. This instrument (Georgiou & Kyza, 2017) was chosen because it has high reliability, all of the items were either developed in Greek or adapted and translated into Greek from published immersion questionnaires (e.g. Cheng et al., 2015; Jennett et al., 2008; Qin et al., 2009) and it was validated with a Greek-speaking population. Examples of statements are the following: "I wanted to spend the time to complete the activity successfully" (measuring interest), "It was easy for me to use the Virtual Reality application" (measuring usability), "I was often excited since I felt as being part of the activity" (measuring emotional attachment).

Data collection

At the first stage of the study, students in both conditions were informed in writing and verbally about the objective of the study by the second author and signed consent forms to voluntarily participate in the study. The study followed APA ethical standards and GDPR guidelines. It meets the ethical guidelines, including adherence to the legal requirements of the country where the study was conducted. In the second stage, students in both conditions completed the vocabulary test as a pre-test. At the third stage of the study, students of the experimental group interacted with the immersive VR application at a specific location in university premises where the equipment was located, with the second author assuming the responsibility of data collection, under the supervision of the first author of the study. The twenty undergraduate students of the experimental group interacted with the application using a head-mounted VR

display (Oculus Rift) for learning Italian as a foreign language for an average duration of 20 min ($M = 20.19$ min, SD = 1.09). Students of the control group interacted with the mobile application using a mobile phone that was given to them by the second author at a second location on university premises. The twenty students of the control group used the mobile application for an average duration of 10 min ($M = 10.36$ min, SD = 0.63). In both conditions, participants had to complete the three introductory scenarios of the application in Italian. At the fourth stage of the study, participants in both conditions completed the vocabulary test as a post-test. The post-test was identical to the pre-test. Participants in both conditions also completed the immersion questionnaire.

Data analysis

Data from pre-tests and post-tests were evaluated by the second author with a maximum possible score of 10. Participants received 1 point for each correct answer written in Italian, 0.5 points for each incomplete answer written in Italian, and 0.25 points for each answer that was written in Greek (participants' native language) but showed that participants understood the question which was posed in Italian. Scores were input in a statistical package (IBM SPSS Statistics 25) for analysis. Descriptive (means and standard deviations) and inferential statistics (paired-samples *t*-tests and independent samples *t*-tests) were used for data analysis. Concerning the use of descriptive and inferential statistics for each Research Question: paired samples *t*-tests were used for answering Research Questions 1 and 2, independent samples *t*-tests were used for answering Research Questions 2 and 4, and descriptive statistics were used for answering Research Question 3.

Data from immersion questionnaires were also input in the same file for analysis. Negatively phrased statements' scores were reversed. Composite scores for each dimension of the immersion questionnaire were then calculated and descriptive and inferential statistics were used. Moreover, a reliability analysis using Cronbach's α was used for the instrument as a whole as well as for each one of the three factors of the instrument. An alpha level of 0.05 was set a priori for all statistical analyses.

Results

Research question 1: the effect of immersive VR apps on vocabulary acquisition in a foreign language

The first research question was the following: "To what extent does an immersive VR application affect the foreign language vocabulary learning of undergraduate students?"

Table 1 shows students' pre-test and post-test vocabulary skills performance for both the experimental and control group.

A paired-samples *t*-test was conducted. Findings revealed a statistically significant difference ($t(19) = -7.71$, $p < 0.01$) in the experimental group students' vocabulary performance when students' pre-test scores ($M = 1.28$, SD = 2.04) were compared to their post-test scores ($M = 3.78$, SD = 2.26).

Table 1. Students' pre-test post-test vocabulary performance for the experimental and control group.

	Pre-intervention		Post-intervention	
	M	SD	*M*	SD
Experimental	1.28	2.04	3.78**	2.25
Control group	1.93	1.99	4.66**	2.45

** = statistically significant result $p < 0.01$.

Research question 2: comparing the effect of an immersive VR application to a mobile application on vocabulary acquisition in a foreign language

The second research question attempted to compare the two versions of the application (immersive VR and mobile application) concerning their effect on vocabulary acquisition in a foreign language. The first step of data analysis consisted of examining group equivalence. Participants were pre-tested concerning their knowledge of the Italian language. There were no statistically significant differences between the two groups when an independent-samples *t*-test was performed ($t(2) = -1.020$, $p = 0.34$) to compare the pre-test of students' vocabulary performance for the experimental group ($M = 1.28$, SD = 2.04) and the control group ($M = 1.93$, SD = 1.99). Therefore, group equivalence was established.

Similarly to the experimental group, the control group also experienced an increase in vocabulary performance (Table 1). A paired-samples *t*-test showed that there was a statistically significant difference (t (19) = −9.79, $p < 0.01$) in the control group students' vocabulary performance when students' pre-test scores ($M = 1.93$, SD = 1.99) were compared to their post-test scores ($M = 4.66$, SD = 2.45).

As both groups experienced statistically significant progress in their scores, an independent samples *t*-test was used to examine whether the control group outperformed the experimental group. Results did not show a statistically significant difference between the two groups' post-test scores (t (2) = −1.179, $p = 0.25$).

Research question 3: evaluation of immersion of VR applications

The third research question was the following: "How do students evaluate their immersive VR learning experience concerning their engagement, engrossment, and total immersion?"

Concerning the validation of the instrument that was used in the present study to measure engagement, engrossment, and total immersion (Georgiou & Kyza, 2017), a reliability analysis, using the data from our study, showed a Cronbach's α of 0.87 when all 21 items were included. Specifically, (a) engagement, which consisted of two factors: interest and usability had a Cronbach's α of 0.79, (b) engrossment, which consisted of two factors: "Emotional Attachment" and "Focus of Attention" had a Cronbach's α of 0.76 and (c) "total immersion", which consisted of two factors: "Presence" and "Flow" had a Cronbach's α of 0.77.

Students' scores concerning evaluating engagement, engrossment, and total immersion of the application for both the experimental and control group are shown in Table 2.

Engagement ($M = 5.96$, SD = 0.79), engrossment ($M = 4.98$, SD = 0.98), and immersion ($M = 4.88$, SD = 1.13) received relatively high scores (higher than 4.8 out of 7) by VR participants.

Table 2. Students' evaluation scores concerning engagement, engrossment, and total immersion of the VR and mobile application.

Dimensions	Experimental group		Control group	
1.Engagement	$M = 5.96$	SD = 0.79	$M = 6.32$	SD = 0.49
1.a. Interest	$M = 5.78$	SD = 0.90	$M = 6.04$	SD = 0.84
1.b. Usability	$M = 6.15$	SD = 0.88	$M = 6.60$*	SD = 0.40
2. Engrossement	$M = 4.98$	SD = 0.99	$M = 5.08$	SD = 0.90
2.a.Emotional attachment	$M = 4.88$	SD = 1.19	$M = 5.02$	SD = 1.04
2.b. Focus of Attention	$M = 5.07$	SD = 1.06	$M = 5.15$	SD = 1.15
3.Total immersion	$M = 4.88$	SD = 1.13	$M = 4.64$	SD = 1.05
3.a. Presence	$M = 4.60$	SD = 1.21	$M = 4.34$	SD = 1.46
3.b. Flow	$M = 5.15$	SD = 1.37	$M = 4.95$	SD = 1.24

* = statistically significant result $p < 0.05$.

Research question 4: comparing the evaluation of an immersive VR application to a mobile application

Control group students also evaluated the mobile application with relatively high scores (higher than 4.6 out of 7) for all three dimensions. Engagement ($M = 6.32$, SD = 0.49), engrossment ($M = 5.08$, SD = 0.90), and immersion ($M = 4.64$, SD = 1.05) received relatively high scores (higher than 4.6 out of 7) by control group participants who used the mobile application (Table 2).

The study did not find a statistically significant difference between the two conditions, when an independent-samples *t*-test was conducted, on any of the dimensions that were examined, except for usability. Concerning usability, the evaluation score of control group participants for the usability of the mobile application ($M = 6.60$, SD = 0.40) was higher than the evaluation score of experimental group participants for the usability of the VR application ($M = 6.15$, SD = 0.88) and this difference was statistically significant ($t\ (38) = -2.08$, $p < 0.05$).

Discussion

This study attempted to answer four research questions using a quasi-experimental pre-test post-test control group design that compared an experimental group of students who interacted with a VR application and a control group of students who used a mobile application. The first research question examined the extent to which undergraduate students' interaction with an immersive VR application can affect their learning concerning vocabulary acquisition in a foreign language. Findings showed a statistically significant increase from pre-test to post-test scores in vocabulary acquisition for the group of students who used the VR application. This finding indicates that immersive VR applications are effective in supporting vocabulary acquisition in a foreign language even through a short use of approximately 20 min per participant, and it is in agreement with the findings of a recent study conducted by Tai et al. (2020), which also supported the effectiveness of VR for vocabulary learning. The contribution of our study lies in that it provided some preliminary empirical data that indicate a potential educational value of VR applications, which is something that was lacking from previous studies that focused on the same topic (Cheng et al., 2017; Kaplan-Rakowski & Wojdynski, 2018; Xie et al., 2019). Specifically, previous studies either did not find statistically significant differences between the conditions they examined (Cheng et al., 2017), or they did not use a quantitative, experimental design (Xie et al., 2019) or they did not measure learning gains (Kaplan-Rakowski & Wojdynski, 2018). However, it is important to acknowledge that due to the recency effect of the pre-test, post-test in the present study, we cannot be certain that the learning gains would have a long-term duration.

The second research question compared an immersive VR application with a mobile application concerning its effect on vocabulary acquisition. Findings showed that both versions of the application were effective, as in both cases a statistically significant increase from pre-test to post-test scores in vocabulary acquisition was observed. Moreover, the two versions of the application were equally effective with respect to increasing vocabulary performance in a foreign language. VR applications therefore do not seem to be superior compared to mobile applications with respect to learning effectiveness, provided that the applications share the same instructional design and are based on the same pedagogical methods. The study adds empirical evidence showing the effectiveness of mobile applications for learning vocabulary in a foreign language, and agrees with a few recent studies that had similar findings in the same domain (Chen et al., 2019; Klimova, 2019).

The third research question asked for students' evaluation of their immersive VR learning experience concerning their engagement, engrossment, and total immersion and showed that the VR application scored high in all three dimensions. This indicates that VR participants were engaged in the application, they showed an emotional attachment and a focus of attention in the application and they felt immersed with respect to presence and flow. The fourth research question attempted

to compare the evaluation of the two versions of the application (VR and mobile application) and showed that both versions received relatively high scores in all three dimensions. The only dimension in which a statistically significant difference was observed between the two versions, in favor of the mobile application, was "usability". This finding indicates that the mobile application was more user-friendly compared to the VR application.

The findings of the second and fourth research questions suggest that the financial investment in the VR version of the application is justified for providing an immersive experience to learners but VR does not necessarily provide a better or more effective learning experience when compared to the mobile version of the application. These findings agree with recent studies (Zhang et al., 2020) that compared VR to more traditional technology-enhanced learning environments and did not find VR to be superior. Zhang et al. (2020) found that there is no significant difference between traditional instructional media and VR for transferring well-structured knowledge. However, this was not the case for transferring ill-structured knowledge, in which case VR was found to be more effective at promoting learning than traditional instructional media (Zhang et al., 2020). Vocabulary skills fall under well-structured knowledge, which might explain why participants using VR applications did not outperform participants using mobile applications. It is possible that if ill-structured knowledge was tested in the present study, results concerning the effectiveness of VR applications might be different.

The present study provides preliminary empirical data indicating the learning effectiveness and capability for engagement of VR applications for language learning, even though VR was not shown to be superior compared to a mobile version of the same application. Examining the effect of VR on language learning in the context of ill-structured knowledge can be a direction for future studies. Future research should also explore the potential educational value of VR applications for both language learning and learning in different disciplines through large scale, longitudinal studies. This suggestion agrees with Parmaxi (2020), who found that most studies in the VR corpus of her systematic review, conducted between 2015 and 2018 employed short-term programs in specific contexts. Parmaxi (2020) therefore suggested that future studies should extend their duration and sample, providing opportunities to understand the long-term effects of VR interventions.

Limitations

The study has several limitations. First of all, convenience sampling, rather than random sampling, was used. The size of the sample was small and the duration of the data collection of the study was limited to two weeks. Students only interacted with the applications once, for a limited time that consisted of 10–20 min. The recency effect of the pre-test and post-test and the lack of a delayed post-test prohibit making any claims of long-term learning gains on language learning. Moreover, the study was solely based on quantitative data. For these reasons, the results of this study have limited generalizability.

Instructional implications for practitioners

The findings of this study indicate that university professors would benefit from having undergraduate students use either VR applications or mobile applications to increase their vocabulary performance when learning a new language. Provided that the VR and mobile applications were found to be equally effective in this study, any one of the two could be used to achieve the same result. Admittedly, the mobile version seemed to have higher usability and achieved the same learning result compared to the VR version in less time. The results of this study also indicate that instructors' integration of both VR and mobile applications in their foreign languages curriculum will most probably result in positive student perceptions concerning their engagement, engrossment, and immersion at least in the context of undergraduate introductory courses to a new language.

Acknowledgments

The authors would like to thank the undergraduate students who voluntarily participated in the study.

Disclosure statement

No potential conflict of interest was reported by the authors(s).

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Appendix

A. Test measuring vocabulary skills with open ended questions in Italian

Please answer the following questions in Italian.

You are in Rome, in a taxi and the taxi-driver asks you the following questions. How do you respond?

1. Come ti chiami? _____
2. Come stai? _____
3. Da dove vieni? _____

You are in Rome, in a hotel. The receptionist asks you the following questions. How do you respond?

4. Che tipo di camera vuole? _____
5. Quante persone? _____
6. Per quanto tempo rimane? _____

You are in Rome, in a restaurant. The waiter asks you the following questions. How do you respond?

7. Cosa desidera? _____
8. Qual cosa da bere? _____
9. Altro? _____
10. Com'è andata la cena? _____

B. Validated questionnaire measuring immersion

Answers are given on a seven-point Likert-type scale, ranging from completely disagree to completely agree.

Engagement [Interest and Usability]

Interest

A1: I liked the activity because it was novel

A2: I liked the type of the activity

A3: I wanted to spend the time to complete the activity successfully
A4: I wanted to spend time to participate in the activity
Usability
A5: It was easy for me to use the Virtual Reality application
A6: I found the Virtual Reality application confusing*
A7: The Virtual Reality application was unnecessarily complex*
A8: I did not have difficulties in controlling the Virtual Reality application
[* The asterisk indicates negatively phrased statements].
Engrossment [Emotional attachment + Focus of attention]
Emotional attachment
B1: I was curious about how the activity would progress
B2: I was often excited since I felt as being part of the activity
B3: I often felt suspense by the activity
Focus of attention
B4: If interrupted, I looked forward to returning to the activity
B5: Everyday thoughts and concerns faded out during the activity
B6: I was more focused on the activity rather on any external distraction
Total immersion [Presence + Flow]
Presence
C1: The activity felt so authentic that it made me think that the virtual characters/objects existed for real
C2: I felt that what I was experiencing was something real, instead of a fictional activity
C3: I was so involved in the activity, that in some cases I wanted to interact with the virtual characters/objects directly
C4: I so was involved, that I felt that my actions could affect the activity
Flow
C5: I didn`t have any irrelevant thoughts or external distractions during the activity
C6: The activity became the unique and only thought occupying my mind
C7: I lost track of time, as if everything just stopped, and the only thing that I could think about was the activity